


Islamic University Of Technology

# OEL Report (Software Part)

**Course No.: EEE 4308**

**Course Name: Digital Electronics Lab**

**Team Name: Silicon 6**

**Students' Names:**

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**Department: EEE**

**Introduction:**

We were asked to design a digital logic circuit for an automatic Tea/Coffee Vending Machine. The system takes a 3-bit input for selecting the beverage and produces a 8 bit output. The first 3 bits represent the beverage code, and the last 5 bits represent the assigned group number. Here, **our group number is 6 so the group number will be 00110 in 5 bits.**

| Beverage type | Input | Output |
|---|---|---|
| Raw Tea | 000 | 100 + Group No. |
| Tea + milk | 001 | 101 + Group No. |
| Tea + sugar | 010 | 110 + Group No. |
| Tea + milk + sugar | 011 | 111 + Group No. |
| Raw coffee | 100 | 001 + Group No. |
| Coffee + milk | 101 | 001 + Group No. |
| Coffee + sugar | 110 | 010 + Group No. |
| Coffee + milk + sugar | 111 | 011 + Group No. |

Additionally, a 1-bit Enable (E) input is included to control the system operation.

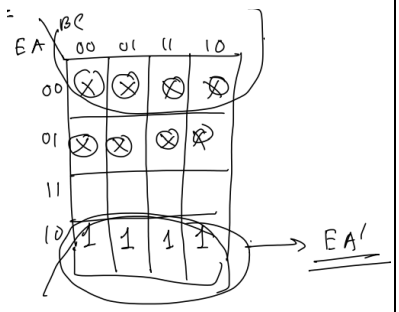
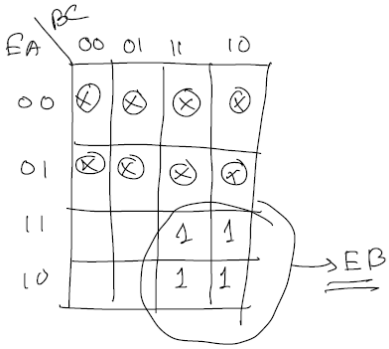
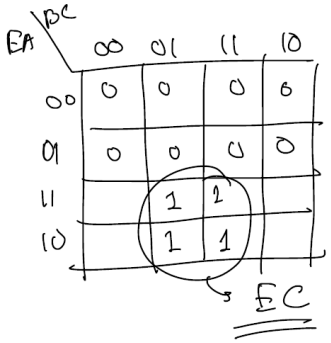
## Description:

We are taking the 3-bit input in the ABC pins and E is out Enable input

**Truth Table:**

| Input | | | | Output | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **E** | **A** | **B** | **C** | **Q0** | **Q1** | **Q2** | **S0** | **S1** | **S2** | **S3** | **S4** |
| 0 | X | X | X | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| | 0 | 0 | 1 | 1 | 0 | 1 | | | | | |
| | 0 | 1 | 0 | 1 | 1 | 0 | | | | | |
| | 0 | 1 | 1 | 1 | 1 | 1 | | | | | |
| | 1 | 0 | 0 | 0 | 0 | 0 | | | | | |
| | 1 | 0 | 1 | 0 | 0 | 1 | | | | | |
| | 1 | 1 | 0 | 0 | 1 | 0 | | | | | |
| | 1 | 1 | 1 | 0 | 1 | 1 | | | | | |

**K-Map**

| Q0 | Q1 | Q2 |
|---|---|---|
|  |  |  |

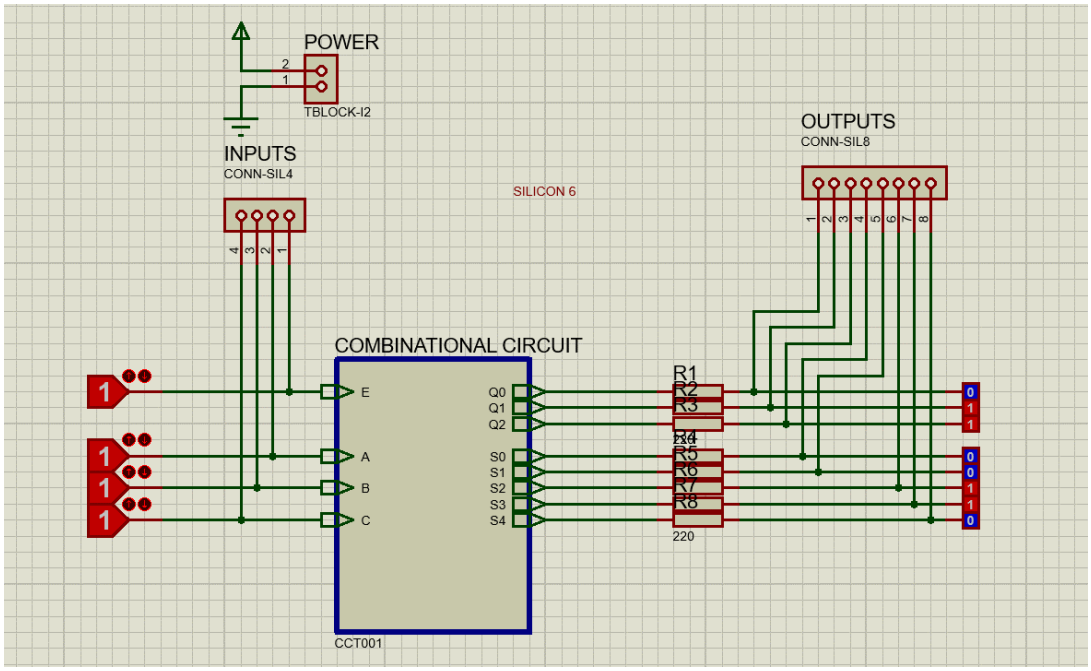
We get the simplified equations:

**Q0 = E.A'; Q1 = E.B; Q2 = E.C**

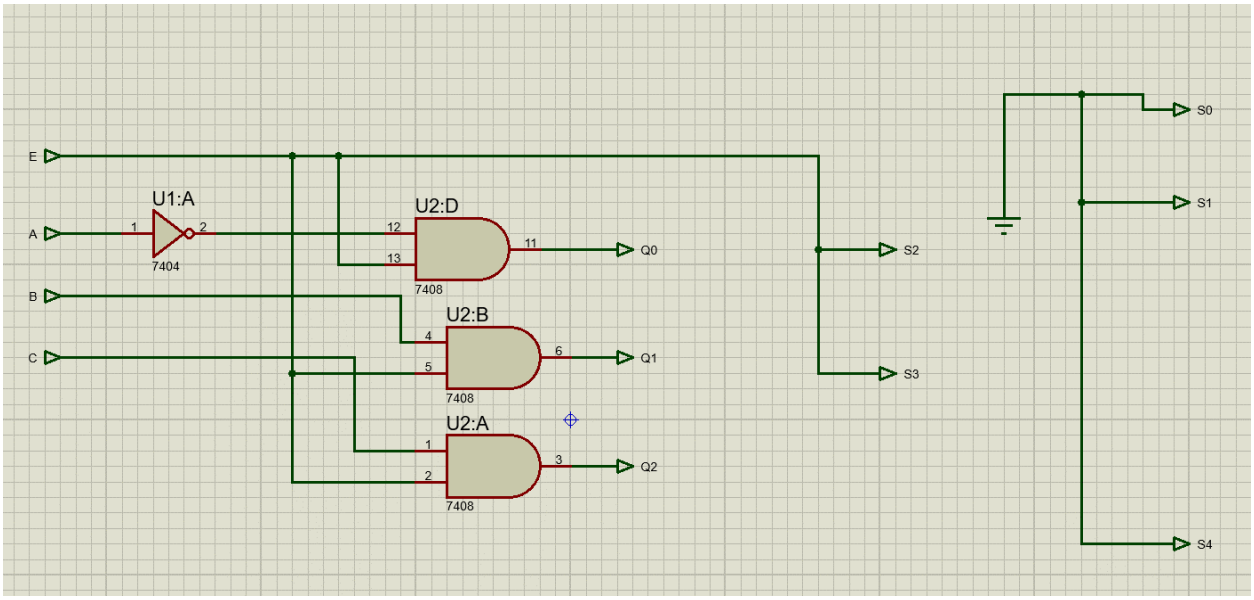
And for the S0-S4 bits, we can AND the Enable pin in this manner:

**S0= 0; S1= 0; S2= E; S3= E; S4=0**

**Circuit Design:**




*Block Diagram of the circuit*




*The combinational Circuit*

**Components used:**

1. 7404: Inverter
2. 7408: AND gate
3. RES: 220-ohm resistors
4. CONN-SIL4, CONN-SIL8: Connector Strip
5. TBLOCK-I2: Terminal Block for Power and Ground

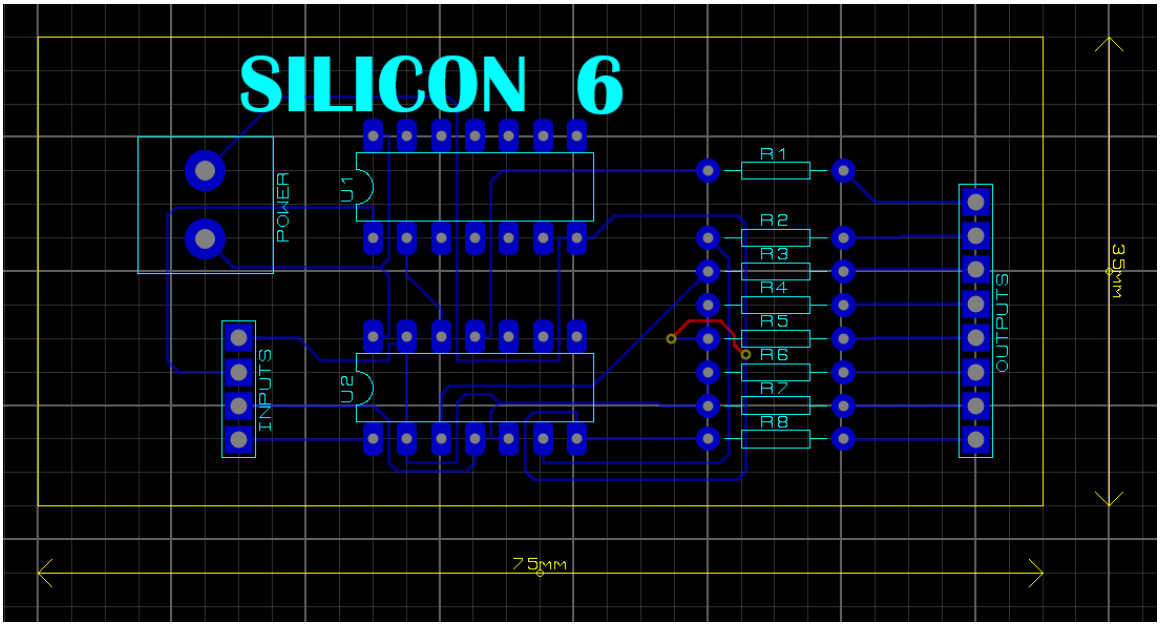
**Combinational Circuit:**

The combinational circuit was designed from our simplified Boolean expressions. For the Q0-Q2 bits, we created the circuit according to their simplified Boolean expression. To incorporate Enable (E), each output was must be ANDed with E. To create our group number in 5 bit, we connected the S2 and S3 and directly to E and S0, S1, and S4 was connected to ground directly since they are always 0 regardless of the input.

After verifying the circuit works with LOGICPROBESs, we designed the PCB.

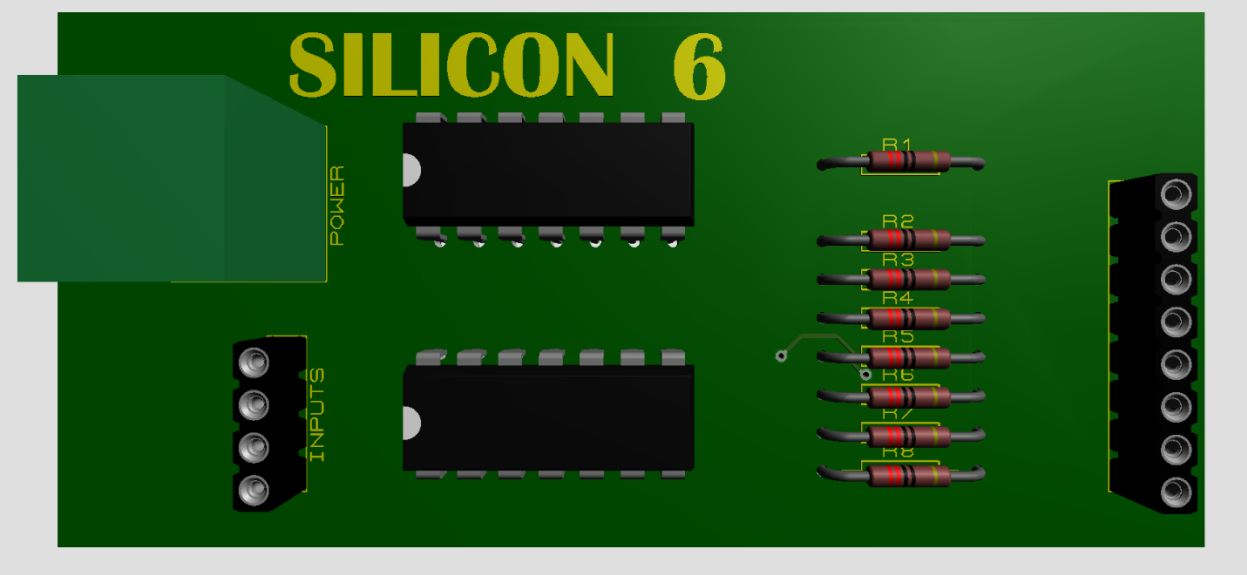
**PCB design:**

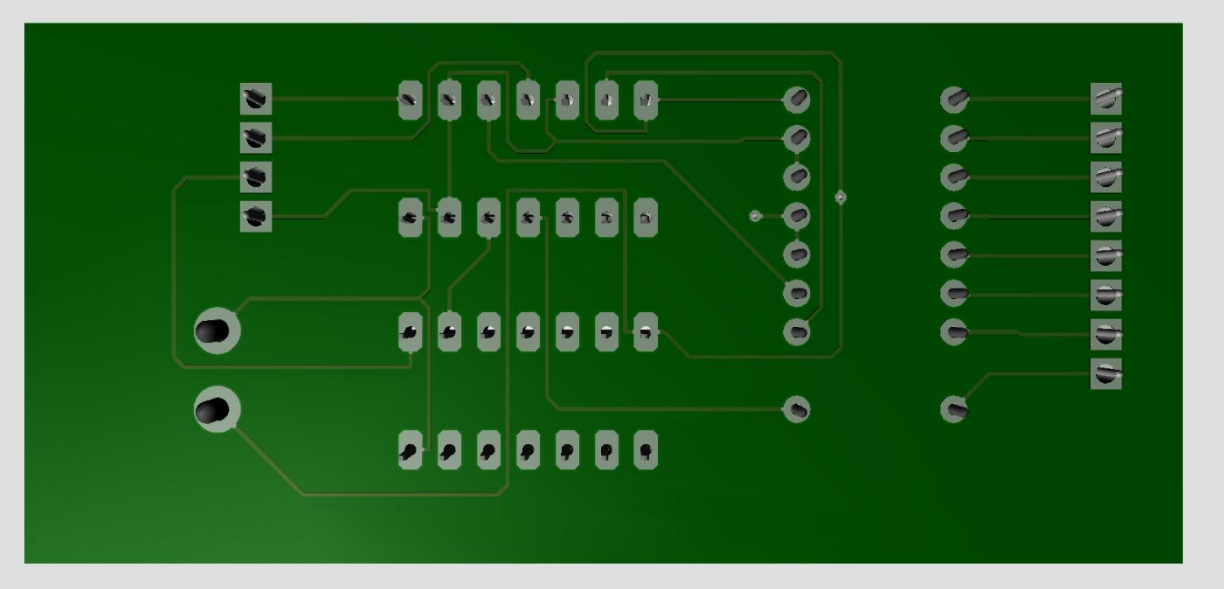
Before designing the PCB, we took two connector strips. A 4-pin connector strip (CONN-SIL4,) for input and 8 pin connector strip for output (CONN-SIL8). We also took a 2-pin screw terminal connector (TBLOCK-I2) for our power and ground.




*PCB Layout*

First, we draw the outline of the area of the PCB in the 'Board Edge' layer. The we placed each of our components in a ordered manner. The 7404 and 7408 were placed along the center, our input connector strip (CONN-SIL4) was placed on the left and the output strip (CONN-SIL8) was placed on the right. Then we used the **auto-routing** for the traces. There were some issues with the auto-routing which we fixed. We ensured there are no copper trace overlapping.



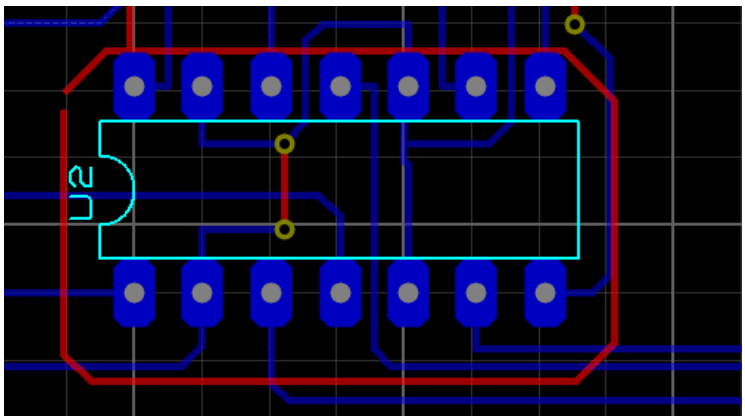
*3d visualization (Top and Bottom)*

The bottom layer contains all the soldering points. The ICs, resistors, and connector strips are mounted in the top layer. The whole PCB layout came out at **75mm x 35 mm** (**2.952 inches x 1.378 inches) .** Which satisfies our design requirement of being within a 4-inch x 4-inch board area.

**Discussion:**

The digital logic circuit and PCB layout for an automatic Tea/Coffee Vending Machine was successfully designed. The schematic was first verified through simulation using logic probes. Then we moved on to the PCB design.

When designing the PCBA, at first the auto-routing was giving the output in a way that was impractical. For example, pins that were supposed to be soldered were given in the top side of the connector strip which would be impossible to use in real life because it would damage the base of the ICs. So, we used a technique where we covered each components outer area with a rectangular trace of the Top Copper (Red color trace) like the figure below first. Then we used the auto routing which gave us proper and practical results. Then we just removed these rectangular traces to get our final PCB design.




*Fixing soldering issue with auto-routing*

We verified the PCB layout is proper because there are **no CRC or DRC errors**. The PCB layout developed falls within the specified 4 inch × 4 inch constraint. The board that we developed was around 3 inch x 1.37 inch. We conclude that the PCB and the logic circuit is working as expected.

**Contributions:**

Schematic building: Ishrak Samin (230021303), Shaheen Rahman Dibya (230021315),

PCB Building and Layout: Ishrak Samin (230021303), Raian Ratul (230021115), Md. Mutassim Al Shahriar Zeem (230021320)

Report Writing: Shaker Abdullah (230021123), Nafis Shahriar Ahmed (230021137), Raian Ratul (230021115)